



# آمار و احتمال مهندسی

## Engineering Probability & Statistics

Lecture Notes ■ جزوه درسی

### Module 8

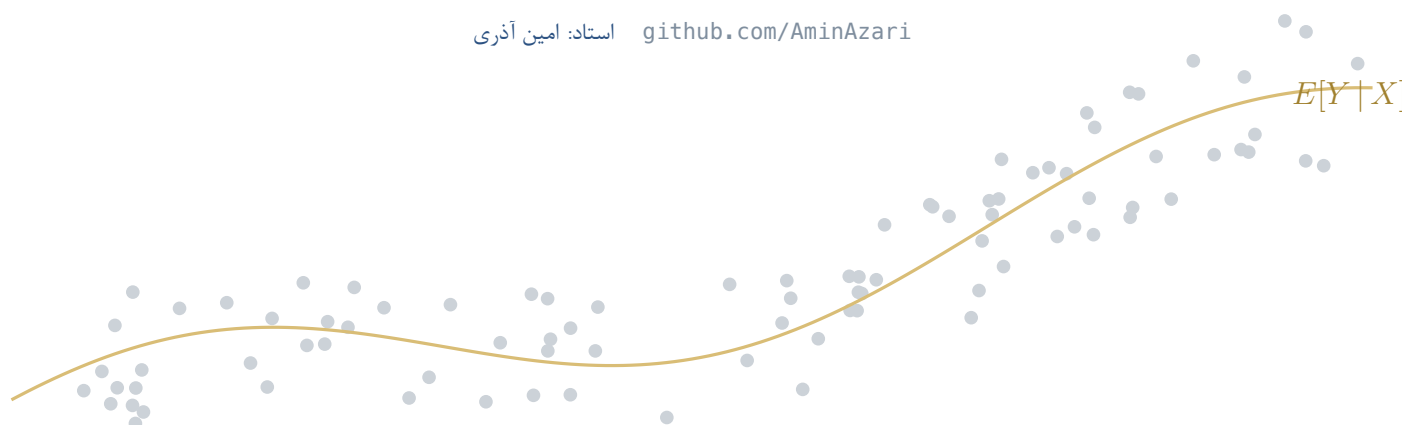
## Conditional Expectation and Variance

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**Learning Objectives**

After completing this module, students will be able to:

1. Compute conditional expectations and variances
2. Apply the Law of Total Expectation
3. Apply the Law of Total Variance
4. Use these tools for engineering analysis across multiple operating conditions

## 8.1 Introduction

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Many engineering systems operate under **different conditions** (channel states, modes, environments). We often need the **overall average performance** computed from conditional performance in each state.

- ▶ What is the average BER when the channel randomly switches between good and bad?
- ▶ What is the overall measurement variance when the sensor operates in different environments?

## 8.2 Conditional Expectation $E[X|Y=y]$

### Definition

#### Continuous:

$$E[X|Y = y] = \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y) dx$$

#### Discrete:

$$E[X|Y = y] = \sum_x x \cdot p_{X|Y}(x|y)$$

### ■ Interpretation

$E[X|Y=y]$  is the **average value of X** when we know  $Y = y$ . It is a function of  $y$ .

### ■ $E[X|Y]$ as a Random Variable

When  $Y$  is itself random,  $E[X|Y]$  is a **random variable** — it takes the value  $E[X|Y=y]$  when  $Y = y$ .

### Engineering Example

Average received power given channel state:

- ▶ Good channel ( $Y=1$ ):  $E[P|Y=1] = 10 \text{ mW}$
- ▶ Poor channel ( $Y=0$ ):  $E[P|Y=0] = 1 \text{ mW}$
- ▶  $P(Y=1) = 0.7$ ,  $P(Y=0) = 0.3$

### 8.3 Conditional Variance $\text{Var}(X|Y=y)$

#### Definition

$$\text{Var}(X|Y = y) = E[(X - E[X|Y = y])^2|Y = y] = E[X^2|Y = y] - (E[X|Y = y])^2$$

#### ■ Interpretation

The conditional variance measures the **remaining uncertainty in X** after Y is known.

#### Engineering Example

Measurement precision depends on temperature:

- ▶ At  $T=20^\circ\text{C}$ :  $\text{Var}(X|T=20) = 0.01$  (high precision)
- ▶ At  $T=80^\circ\text{C}$ :  $\text{Var}(X|T=80) = 0.09$  (low precision due to thermal effects)

## 8.4 Law of Total Expectation (Tower Property)

### ■ Statement

$$E[X] = E[E[X|Y]]$$

**Discrete version:**

$$E[X] = \sum_y E[X|Y=y] \cdot P(Y=y)$$

**Continuous version:**

$$E[X] = \int_{-\infty}^{\infty} E[X|Y=y] \cdot f_Y(y) dy$$

### ■ Proof (Discrete)

$$\begin{aligned} E[E[X|Y]] &= \sum_y E[X|Y=y] \cdot P(Y=y) \\ &= \sum_y [\sum_x x \cdot P(X=x|Y=y)] \cdot P(Y=y) \\ &= \sum_x \sum_y x \cdot P(X=x, Y=y) \\ &= \sum_x x \cdot P(X=x) = E[X] \quad \checkmark \end{aligned}$$

### ■ Interpretation

The **overall average** equals the **weighted average of conditional averages**, weighted by the probability of each condition.

#### Engineering Example: Average BER Across Channel States

A wireless system switches between three modes:

- ▶ Mode 1 (60% of time):  $\text{BER} = 10^{-6}$
- ▶ Mode 2 (30% of time):  $\text{BER} = 10^{-4}$
- ▶ Mode 3 (10% of time):  $\text{BER} = 10^{-2}$

$$\begin{aligned} \text{Overall BER} &= E[\text{BER}] = 0.6(10^{-6}) + 0.3(10^{-4}) + 0.1(10^{-2}) \\ &= 0.6 \times 10^{-6} + 30 \times 10^{-6} + 10000 \times 10^{-6} \approx 1.003 \times 10^{-3} \end{aligned}$$

**Engineering insight:** The overall BER is dominated by the worst mode, even though it occurs only 10% of the time.

#### Engineering Example: Average Lifetime with Defect Types

Components are defect-free (90%) or have a hidden defect (10%):

- ▶ Defect-free:  $E[T|\text{good}] = 10000$  hours
- ▶ With defect:  $E[T|\text{defect}] = 500$  hours

$$E[T] = 0.9(10000) + 0.1(500) = 9050 \text{ hours}$$

## 8.5 Law of Total Variance

### ■ Statement

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

| Term                 | Name                          | Interpretation                     |
|----------------------|-------------------------------|------------------------------------|
| $\text{Var}(X)$      | Total variance                | Overall uncertainty in X           |
| $E[\text{Var}(X Y)]$ | Expected conditional variance | Average “within-group” variability |
| $\text{Var}(E[X Y])$ | Variance of conditional means | “Between-group” variability        |

### ■ Proof

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = E[E[X^2|Y]] \text{ (tower property)}$$

$$= E[\text{Var}(X|Y) + (E[X|Y])^2]$$

$$= E[\text{Var}(X|Y)] + E[(E[X|Y])^2]$$

$$(E[X])^2 = (E[E[X|Y]])^2$$

$$\text{Var}(X) = E[\text{Var}(X|Y)] + E[(E[X|Y])^2] - (E[E[X|Y]])^2$$

$$= E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]) \quad \checkmark$$

### ■ Interpretation: Decomposition of Uncertainty

Total uncertainty = Average uncertainty within each condition + Variability of the mean across conditions

#### Engineering Example: Sensor in Two Environments

A sensor operates indoors ( $Y=1$ , prob 0.6) and outdoors ( $Y=2$ , prob 0.4):

- Indoor:  $E[X|Y=1] = 25^\circ\text{C}$ ,  $\text{Var}(X|Y=1) = 1$
- Outdoor:  $E[X|Y=2] = 15^\circ\text{C}$ ,  $\text{Var}(X|Y=2) = 16$

$$E[\text{Var}(X|Y)] = 0.6(1) + 0.4(16) = 7.0 \text{ (average within-condition variance)}$$

$$\text{Var}(E[X|Y]) = E[(E[X|Y])^2] - (E[E[X|Y]])^2$$

$$= 0.6(25^2) + 0.4(15^2) - [0.6(25) + 0.4(15)]^2 = 375 + 90 - (21)^2 = 465 - 441 = 24$$

$$\text{Var}(X) = 7.0 + 24.0 = 31.0$$

Most variance comes from the **between-condition** difference (24 out of 31).

## 8.6 MATLAB Examples

### Example 1: Law of Total Expectation

```

%% Average BER across channel states
% State probabilities and conditional BERs
p_state = [0.6, 0.3, 0.1];
BER_cond = [1e-6, 1e-4, 1e-2];

% Total expectation
BER_overall = sum(p_state .* BER_cond);
fprintf('Overall BER = %.4e\n', BER_overall);

% Simulation verification
N = 1000000;
states = randsample(1:3, N, true, p_state);
errors = rand(1,N) < BER_cond(states);
BER_sim = mean(errors);
fprintf('Simulated BER = %.4e\n', BER_sim);

```

MATLAB

### Example 2: Law of Total Variance

```

%% Total Variance Decomposition: Sensor Example
N = 100000;
p_indoor = 0.6;

% Generate environment (1=indoor, 2=outdoor)
env = (rand(1,N) > p_indoor) + 1;

% Generate measurements conditional on environment
X = zeros(1, N);
X(env==1) = 25 + 1*randn(1, sum(env==1)); % Indoor: N(25, 1)
X(env==2) = 15 + 4*randn(1, sum(env==2)); % Outdoor: N(15, 16)

% Total variance
total_var = var(X);

% E[Var(X|Y)]
within_var = p_indoor*var(X(env==1)) + (1-p_indoor)*var(X(env==2));

% Var(E[X|Y])
cond_means = [mean(X(env==1)), mean(X(env==2))];
between_var = p_indoor*cond_means(1)^2 + (1-p_indoor)*cond_means(2)^2 ...
    - (p_indoor*cond_means(1) + (1-p_indoor)*cond_means(2))^2;

fprintf('Total Var(X) = %.2f (theory: 31.0)\n', total_var);
fprintf('E[Var(X|Y)] = %.2f (theory: 7.0)\n', within_var);
fprintf('Var(E[X|Y]) = %.2f (theory: 24.0)\n', between_var);
fprintf('Sum          = %.2f\n', within_var + between_var);

```

MATLAB

**Example 3: Conditional Expectation as Best Predictor**

MATLAB

```

%% E[X|Y] as a function of Y
% Joint: X = Y + N, Y ~ Uniform(0,10), N ~ N(0, 1)
N_samp = 50000;
Y = 10*rand(1, N_samp);
X = Y + randn(1, N_samp);

% Estimate E[X|Y=y] for various y
y_bins = 0:0.5:10;
E_X_given_Y = zeros(size(y_bins)-[0 1]);
for i = 1:length(y_bins)-1
    mask = Y >= y_bins(i) & Y < y_bins(i+1);
    E_X_given_Y(i) = mean(X(mask));
end
y_centers = (y_bins(1:end-1) + y_bins(2:end))/2;

figure;
scatter(Y(1:5000), X(1:5000), 1, 'b', '.');
hold on;
plot(y_centers, E_X_given_Y, 'r-', 'LineWidth', 3);
plot(y_centers, y_centers, 'g--', 'LineWidth', 2);
xlabel('Y'); ylabel('X');
legend('Data (X,Y)', 'E[X|Y] (estimated)', 'E[X|Y] = Y (theory)');
title('Conditional Expectation');

```

## 8.7 Practice Problems

### Problem 1

A digital link operates over two channel types:

- ▶ Type A (prob 0.7):  $E[\text{delay}|A] = 5\text{ms}$ ,  $\text{Var}(\text{delay}|A) = 2$
- ▶ Type B (prob 0.3):  $E[\text{delay}|B] = 20\text{ms}$ ,  $\text{Var}(\text{delay}|B) = 25$

Find (a)  $E[\text{delay}]$ , (b)  $\text{Var}(\text{delay})$ .

#### Solution

- (a)  $E[D] = 0.7(5) + 0.3(20) = 3.5 + 6 = 9.5 \text{ ms}$   
 (b)  $E[\text{Var}(D|\text{type})] = 0.7(2) + 0.3(25) = 1.4 + 7.5 = 8.9$   
 $\text{Var}(E[D|\text{type}]) = 0.7(5^2) + 0.3(20^2) - 9.5^2 = 17.5 + 120 - 90.25 = 47.25$   
 $\text{Var}(D) = 8.9 + 47.25 = 56.15$

### Problem 2

A random number  $N$  of components are tested.  $N \sim \text{Poisson}(10)$ . Each has independent failure probability  $p=0.05$ . Let  $X$  = number of failures. Find  $E[X]$  and  $\text{Var}(X)$ .

**Solution**  $X|N \sim \text{Binomial}(N, 0.05)$ .  $E[X|N] = 0.05N$ .  $\text{Var}(X|N) = N(0.05)(0.95) = 0.0475N$ .

$E[X] = E[E[X|N]] = E[0.05N] = 0.05(10) = 0.5$

$\text{Var}(X) = E[\text{Var}(X|N)] + \text{Var}(E[X|N]) = E[0.0475N] + \text{Var}(0.05N) = 0.0475(10) + 0.05^2(10) = 0.475 + 0.025 = 0.5$

### Problem 3

Write MATLAB code to verify Problem 2 by simulation.

#### Solution

```
N_trials = 100000; p = 0.05; lambda = 10;
N = poissrnd(lambda, 1, N_trials);
X = binornd(N, p);
fprintf('E[X]: theory=0.5, sim=%.3f\n', mean(X));
fprintf('Var(X): theory=0.5, sim=%.3f\n', var(X));
```

MATLAB

*Next Module: Module 9 — Covariance and Correlation*